

SPADEADAM, United Kingdom

WMO: 031650

Lat: 55.0503N

Lon: 2.5544W

Elev: 1066

StdP: 14.14

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.8	(c) 25.0	(d) 19.7	(e) 15.4	(f) 24.6	(g) 22.7	(h) 17.8	(i) 28.1	(j) 37.6	(k) 45.6	(l) 33.9	(m) 43.3	(n) 6.8	(o) 260	(p) 0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	71.8	61.7	68.2	59.6	65.8	58.5	63.4	69.5	61.6	66.4	59.9	63.7	10.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	83.6	64.8	59.4	78.5	62.6	58.6	76.5	61.7	29.3	70.1	27.9	65.9	26.7	64.1	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.2	27.6	24.7	18.7	78.9	4.0	4.1	15.8	81.9	13.5	84.3	11.3	86.6	8.4	89.6		
			17.9	67.2	3.9	2.7	15.1	69.1	12.8	70.7	10.6	72.2	7.8	74.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	36.6	36.9	39.0	43.2	48.5	53.3	57.0	55.5	52.3	47.5	40.8	37.4
	DBStd	8.82	5.23	5.12	4.93	5.04	5.40	4.60	4.40	3.33	4.23	4.85	5.38	6.23
	HDD50	2244	414	366	343	212	93	17	0	2	23	105	279	390
	HDD65	7041	879	786	807	654	511	350	252	294	382	544	727	855
	CDD50	681	0	0	1	7	47	117	217	173	91	26	2	0
	CDD65	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH74	52	0	0	0	0	5	9	34	3	1	0	0	0
	CDH80	4	0	0	0	0	0	1	3	0	0	0	0	0
	WSAvg	11.7	12.9	12.5	12.8	11.6	11.9	10.5	10.1	10.4	11.1	11.7	12.2	12.7
	PrecAvg	48.0	5.3	4.2	3.1	3.0	3.1	3.1	3.7	3.9	3.4	4.7	4.8	5.4
(15) Precipitation	PrecMax	61.0	9.2	10.7	6.0	5.7	5.5	7.8	6.6	8.6	5.6	8.9	12.1	14.3
	PrecMin	33.7	0.5	0.9	0.9	1.2	0.7	0.7	1.4	0.7	0.4	1.1	2.2	1.3
	PrecStd	7.4	2.2	2.7	1.2	1.2	1.4	1.7	1.5	1.9	1.3	2.1	2.2	2.8
	PrecStd	7.4	2.2	2.7	1.2	1.2	1.4	1.7	1.5	1.9	1.3	2.1	2.2	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	52.0	57.2	66.2	72.0	74.8	78.8	70.0	69.8	61.1	55.5	52.2
		MCWB	49.3	45.9	47.8	54.5	59.8	64.1	65.1	61.8	61.0	56.5	54.4	50.2
	2%	DB	48.1	48.3	51.4	58.8	67.9	69.7	73.2	66.4	64.5	58.7	52.2	50.2
		MCWB	47.7	46.6	47.1	50.5	57.0	59.8	63.0	59.7	58.7	55.7	51.0	48.8
	5%	DB	46.3	46.5	48.2	55.1	62.4	66.0	69.6	64.3	61.2	56.8	50.4	48.3
		MCWB	45.9	45.4	45.6	48.8	54.1	58.0	61.5	58.8	56.8	54.7	49.2	47.7
	10%	DB	44.5	44.7	46.3	51.8	57.6	62.4	65.8	62.3	59.2	55.1	48.5	46.4
		MCWB	44.0	43.6	44.7	47.5	51.4	55.8	59.4	58.0	55.7	53.1	47.7	45.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	48.6	50.9	55.1	62.2	64.6	67.2	63.7	63.1	58.6	54.7	51.6
		MCDB	51.0	49.8	56.7	63.5	71.1	71.2	75.9	67.4	67.7	60.6	55.3	52.5
	2%	WB	47.6	46.8	48.0	52.3	57.8	61.3	64.1	61.8	60.0	56.7	52.0	49.9
		MCDB	47.9	47.5	49.4	57.5	65.3	67.7	70.9	64.6	62.9	57.9	52.7	50.5
	5%	WB	45.5	45.4	46.2	50.1	55.2	58.7	62.2	60.3	58.1	55.1	50.0	47.9
		MCDB	45.7	46.1	47.5	53.8	60.8	64.1	68.1	62.8	60.5	56.4	50.7	48.3
	10%	WB	43.7	43.3	44.5	48.1	53.1	56.9	60.2	58.8	56.7	53.3	48.1	45.9
		MCDB	44.1	44.2	45.9	51.0	57.4	61.3	64.6	60.9	59.0	54.7	48.7	46.2
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.8	10.8	13.8	15.2	14.7	14.3	12.3	12.2	10.3	9.0	8.3
		MCDBR	8.7	11.0	13.7	19.7	22.3	21.0	20.4	15.5	15.2	11.7	9.8	10.1
		MCWBR	8.9	9.3	9.9	12.4	13.1	12.5	11.3	9.7	9.9	9.2	9.5	10.1
	5% WB	MCDBR	8.7	9.5	11.8	16.8	19.6	18.4	17.9	12.9	12.8	10.5	9.7	10.0
		MCWBR	9.1	8.9	9.7	11.6	12.3	12.1	11.0	9.5	9.7	9.5	9.9	10.1
			(d) 0.309	(e) 0.322	(f) 0.351	(g) 0.371	(h) 0.372	(i) 0.371	(j) 0.371	(k) 0.361	(l) 0.344	(m) 0.330	(n) 0.305	(o) 0.292
(40) Clear-Sky Solar Irradiance	taub		2.277	2.369	2.315	2.286	2.324	2.350	2.351	2.393	2.447	2.445	2.353	2.249
	Ebn at Noon		195	241	259	270	276	277	275	271	260	235	197	172
	Edh at Noon		18	23	31	36	37	37	36	32	27	22	16	15
	RadAvg		174	380	720	1158	1510	1490	1438	1161	829	451	223	131
(45) All-Sky Solar Radiation	RadStd		24	40	72	122	115	147	156	89	70	55	22	12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page